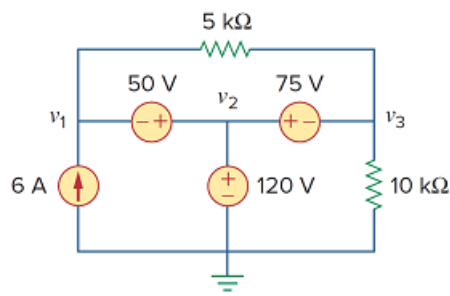


Problem

Obtain the node voltages v_1 , v_2 , and v_3 in the circuit of Fig. 3.81.

Figure 3.81:



Step-by-step solution

Step 1 of 2

Refer to Figure 3.81 in the text book.

Notice that the value of the voltage v_2 is, 120 V .

There is a voltage source exists between two nodes v_1 and v_2 . Hence, the nodes v_1 and v_2 forms a super node. Write the Kirchhoff's voltage law to the super node formed by the voltages v_1 and v_2 .

$$v_2 - v_1 = 50$$

$$v_1 = v_2 - 50$$

Substitute 120 V for v_2 .

$$v_1 = 120 - 50$$

$$= 70\text{ V}$$

Therefore, the value of the voltage v_1 is, 70 V .

Step 2 of 2

Similarly, there is a voltage source exists between two nodes v_2 and v_3 . Hence, the nodes v_2 and v_3 forms a super node. Write the Kirchhoff's voltage law to the super node formed by the voltages v_2 and v_3 .

$$v_2 - v_3 = 75$$

$$v_3 = v_2 - 75$$

Substitute 120 V for v_2 .

$$v_3 = 120 - 75$$

$$= 45\text{ V}$$

Therefore, the value of the voltage v_3 is, 45 V .

Therefore, the values of node voltages v_1 , v_2 and v_3 are 70 V , 120 V and 45 V respectively.